

HTS Master Classifier

NLP-Based Product Classification for International Trade

Milestone #2

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Group 4

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Define Objectives

This project aims to develop an NLP-based multiclass classification system that predicts Harmonized Tariff Schedule (HTS) codes from unstructured product descriptions found in trade documents. By reducing manual research, the system supports trade compliance workflows and helps reduce risks such as shipment delays, incorrect tariff calculations, and penalties caused by inaccurate classification. For this milestone, the project focuses on predicting 4-digit HTS headings, which provide a practical balance between classification detail and data availability.

The core NLP task is supervised multiclass text classification. The input is a natural-language product description, and the output is a predicted 4-digit HTS heading. Supporting tasks include text preprocessing, HTS code extraction, label encoding, TF-IDF feature extraction, model training, model evaluation, and inference.

1. HTS prediction - Predict the correct 4-digit HTS heading from product descriptions.
2. Model comparison - Evaluate multiple NLP and machine learning approaches.
3. Model implementation - Compare Logistic Regression, Linear SVM, Naive Bayes, and a Feedforward Neural Network.
4. Reusable pipeline - Build a pipeline that can classify new product descriptions.
5. Future OCR integration - Support future use with OCR-extracted text from invoices and trade documents.
6. HTS validation - Support future mapping of predictions against the official HTS reference structure.

Literature Review

Current research and best practices in NLP-based classification generally include two practical approaches for this project: traditional machine learning models and neural network models.

Traditional Machine Learning

Traditional machine learning models like Logistic Regression, SVM, Naive Bayes, and Random Forest use TF-IDF features or n-grams to convert text into numerical inputs. These methods are highly relevant to HTS classification because product descriptions contain direct classification signals like “cotton” or “steel”.

TF-IDF weights important, descriptive terms higher. Logistic Regression offers a fast, interpretable baseline, and Linear SVM excels with high-dimensional sparse text data. Naive Bayes is also fast and simple but can struggle with class imbalance or overlapping vocabulary across categories.

Neural Network Models

Neural networks learn flexible relationships between input features and target labels. In this project, the Feedforward Neural Network (FFNN) uses a TF-IDF feature matrix instead of word embeddings, incorporating a hidden layer, ReLU activation, dropout regularization, and an output layer to predict encoded HTS classes.

The FFNN models nonlinear patterns that traditional linear classifiers miss, though it requires more resources and tuning. Trained in PyTorch using dense TF-IDF tensors, 256 hidden units, 0.3 dropout, Adam optimization, and cross-entropy loss, the FFNN performed well but did not outperform the Linear SVM.

Benchmarking

The benchmark compares four completed models: Logistic Regression, Linear SVM, Naive Bayes, and FFNN.

Model	Method	Advantages	Disadvantages
Model 1	Logistic Regression + TF-IDF	Fast, interpretable, strong baseline	Lower performance than SVM and FFNN
Model 2	Linear SVM + TF-IDF	Strong performance with sparse text features	Less naturally probabilistic; may need calibration
Model 3	Naive Bayes + TF-IDF	Very fast, simple, useful baseline	Weakest performance due to class imbalance and overlapping vocabulary
Model 4	FFNN + TF-IDF	Captures nonlinear patterns and performs well	More computationally expensive and requires more tuning

Results of Model Evaluation

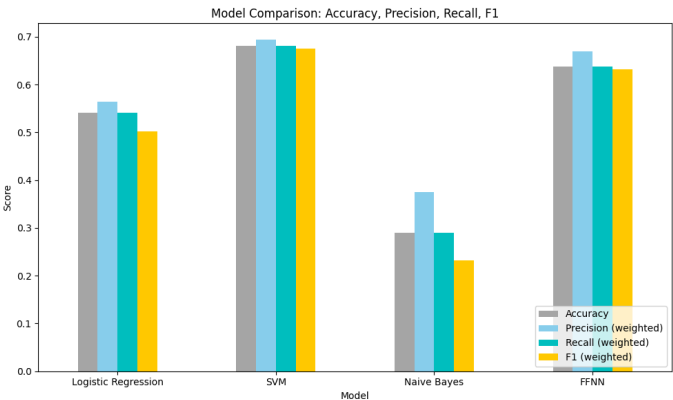
Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Logistic Regression	54.1%	56.4%	54.1%	50.1%
Linear SVM	68.1%	69.4%	68.1%	67.6%
Naive Bayes	28.9%	37.5%	28.9%	23.2%
FFNN	63.8%	66.9%	63.8%	63.3%

The team evaluated four completed models using the same train-test split and TF-IDF feature representation, allowing a consistent comparison across accuracy, weighted precision, weighted recall, and weighted F1-score. Overall, SVM is the best-performing model, achieving the highest accuracy and weighted recall at 68.1%, weighted precision at 69.4%, and weighted F1-score at 67.6%. This suggests that SVM is well suited for HTS classification because product descriptions contain many keyword-based signals, and SVM performs effectively with high-dimensional sparse TF-IDF features.

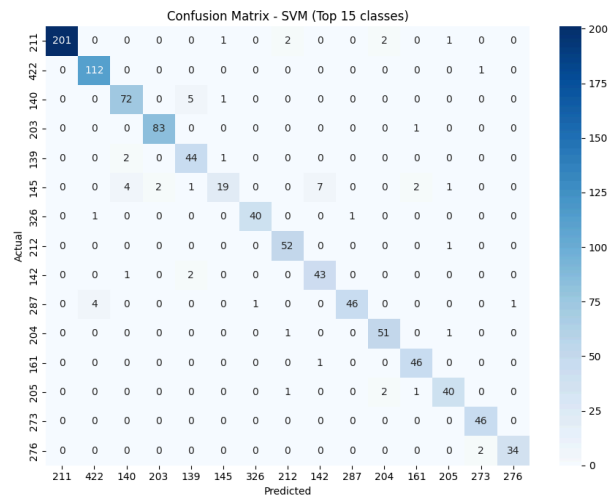
The FFNN is the second-best model, with a weighted F1-score of 63.3%. Although its training loss decreased over 100 epochs, it did not generalize as well as SVM on the test set. Logistic Regression remains a useful fast and interpretable baseline, with a weighted F1-score of 50.1%. Naive Bayes performed weakest, with a weighted F1-score of 23.2%, likely due to class imbalance, many HTS classes, and overlapping product vocabulary across headings.

The evaluation also validates the internal end-to-end modeling workflow. It successfully loads and parses the data, cleans product descriptions, extracts HTS labels, converts codes into 4-digit headings, filters rare classes, vectorizes text with TF-IDF, trains four models, generates predictions, and compares performance. The comparison table, bar chart, and Linear SVM confusion matrix provide a view of model performance, supporting SVM as the current best model for this milestone.

Visualization 1: Model Comparison Accuracy, Precision Recall, F1



Visualization 2: Confusion Matrix



Preliminary Experiments

Preliminary experiments confirm that NLP can successfully predict 4-digit HTS headings from product descriptions. The completed workflow covers data loading, parsing, text cleaning, HTS extraction, label encoding, rare-class filtering, stratified train-test splitting, TF-IDF vectorization, and training four models: Logistic Regression, Linear SVM, Naive Bayes, and FFNN.

Since the Hugging Face source is private, the pipeline uses a local CROSS rulings dataset, loading 18,254 records from the `messages` column via `load_cross_dataset()` against the `hts_2026_revision_10_csv` taxonomy. This field is parsed into `product_description`, `hts_codes`, and `reasoning`, then processed via `clean_text()` to strip redundant prompt language while retaining core attributes like material, chemical identifiers, country, and use.

Using `extract_fields()`, HTS codes are extracted to establish target labels for the descriptions. The notebook isolates the first four digits of the initial listed HTS code without decimals to create the `hts_4digit` label, which is numerically encoded via `LabelEncoder`.

To counter data sparsity, classes with fewer than 10 samples are removed, refining the original 18,254 records across 549 unique classes down to 17,378 records across 258 classes. This data is then divided using an 80/20 stratified split to preserve class proportions.

TF-IDF vectorization converts descriptions into numerical features using unigrams, bigrams, a maximum of 50,000 features, and `min_df=2`, resulting in a final feature space of 27,534 predictive terms and phrases.

All models are evaluated on this shared feature space. Linear SVM performs best (68.1% accuracy, 69.4% precision, 68.1% recall, 67.6% F1), followed by the FFNN (63.8% accuracy, 66.9% precision, 63.8% recall, 63.3% F1), Logistic Regression (54.1% accuracy, 56.4% precision, 54.1% recall, 50.1% F1), and Naive Bayes (28.9% accuracy, 23.2% F1).

The evaluation confirms the models functional capacity to generate predictions, evaluate performance metrics, visualize comparative success, and construct a confusion matrix for the top-performing Linear SVM model.